

JAVA REASONING QUICK CHEATSHEET

LAST-MINUTE PLACEMENT EXAM REVISION (1-2 MIN SCAN)



BASIC SYNTAX & DATATYPES



- Variables `int v = 10;`
- `System.out.println("Hello");`
- Primitive Types
`int, double, char, boolean`
- Reference Types



STRING HANDLING



- String creation `s = "example"`
- `length()`
- `equals()`
- `substring(a, b)`
- `toUpperCase()`
- `trim()`



CONTROL FLOW (If/Else, Loops)



- If-else logic structures
- for loop, while loops
- switch case
- break, continue

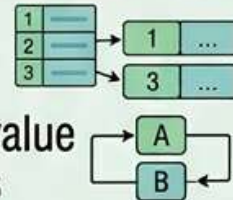
if elif {
continue:
elseif



ARRAYS & COLLECTIONS



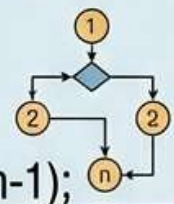
- Array creation `int[] a = {1,2,3};`
- ArrayList
- HashMap
- HashSet: key-value
key-value pairs



FUNCTIONS & RECURSION



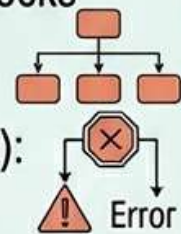
- Function declaration
(public, static, return type)
- call `greet("name")`
- Simple recursive:
`if(n<=1) return 1;`
`else return n*fact(n-1);`



EXCEPTION HANDLING



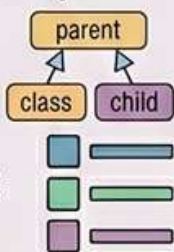
- try, catch, finally blocks
- throw exceptions
management
- `def test_exception(x):`
try:
`y=1/x`
except: return 'Error'



OOP CONCEPTS (Inheritance, Polymorphism)



- class syntax
 - objects
 - super
 - self
- method overriding
- `def test_decorator(func):`
wrap():
`func() wrap()`
wrap()
- list comprehensions
`[x for x in list]`



BUILT-IN LIBRARIES & INTERFACES



- common Math class
- `max()`
- `min()`
- `abs()`
- `random()`
- common interfaces
(Serializable, Comparable)
- random interfaces
- random list
and interface

